

Lesson 11 · AWS Bedrock Agents

PDF: [this lesson](#) · **Install (Linux · macOS · Windows):** [guide](#) · [PDF](#)

Part of [local-ai-lab](#) — a hands-on course for building local AI.

Course home: <https://nikolareljin.github.io/local-ai-lab/> **Source:** <https://github.com/nikolareljin/local-ai-lab>

Lessons: [1 · RAG](#) → [2 · MCP](#) → [3 · Hybrid retrieval](#) → [4 · RAG safety](#) → [5 · RAG evaluation](#) → [6 · Repo assistant](#) → [7 · LangChain](#) → [8 · LangGraph](#) → [9 · Ollama tools](#) → [10 · Semantic Kernel](#) → **11 · Bedrock Agents (you are here)** → [12 · Google ADK](#) → ... → [15 · Docs from changes](#)

Status: planned. Outline below; full published slideshow lesson + step-by-step coming later. This is a **managed cloud** lesson — you build and drive it from a **local** dev environment (AWS CLI/SDK) against Amazon Bedrock. [□](#) the repo to follow along.

The idea

You built RAG and agents by hand. **Amazon Bedrock Agents** is the **managed** version: a hosted agent that combines a **knowledge base** (managed RAG over your documents) with **action groups** (tools backed by Lambda). This lesson shows how the primitives you wrote map onto a cloud agent platform — and where the trade-offs are (less control, less ops).

What you'll learn

- **Bedrock model access** — enabling foundation models (Claude, Titan, Llama) in your account
- **Knowledge Bases** — managed ingestion + embeddings + vector store over your documents/ (this is Lesson 1's pipeline, as a service)
- **Action groups** — defining tools with an OpenAPI schema, implemented by a Lambda (the cloud equivalent of Lesson 5's `search_docs`)
- **Agent orchestration** — how Bedrock plans, retrieves, and calls actions; reading the trace
- **Driving it locally** — invoking the agent from your machine with `boto3` / AWS CLI

The design we'll build

```
import boto3

agent = boto3.client("bedrock-agent-runtime") # run locally, talks to AWS

resp = agent.invoke_agent(
    agentId="XXXX", agentAliasId="YYYY",
    sessionId="demo-1",
    inputText="How do I reset the device? Cite the manual.",
)
# stream the completion chunks; the agent transparently used the Knowledge Base
# (managed RAG) and any action groups, then grounded its answer in your docs.
```

The mapping: Knowledge Base = Lesson 1 retrieval (managed). Action group + Lambda = Lesson 9 tool calling (managed). Agent = Lesson 8's orchestration (managed). Same ideas, someone else runs the servers — you trade transparency and cost for less ops.

Builds on

Concept	From	Cloud equivalent
extract → chunk → embed → retrieve	Lesson 1	Bedrock Knowledge Base
a tool the agent can call	Lesson 9	Action group + Lambda
orchestration / control flow	Lesson 8	Bedrock Agent runtime

Prerequisites

An AWS account with Bedrock model access, the AWS CLI configured locally, and `boto3`. Lessons [1](#) and [9](#) give the mental model.

Next lesson

[Lesson 12 · Google AI Development Kit →](#)

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